IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion

Executive Summary

- Summary of methodologies
- First, we collected the date from SpaceX
- Second, we conducted an EDA on the Data
- Third, we make Folium maps to showcase the data in more representing way
- Fourth, we applied feature selection and normalization to prepare data for modeling
- Fifth, we trained and evaluated machine learning models using skit-learn

Introduction

- Space Y is a new rocket company aiming to compete with SpaceX, founded by (Allon Musk). As a data scientist, my role is to analyze SpaceX launch data to guide Space Y’s decisions on launch pricing and first-stage reuse. Using dashboards and data-driven insights, this project seeks to answer key operational and financial questions without relying on complex rocket science calculations.

Section 1

Methodology

Methodology

- Executive Summary
- Data collection methodology:
  - We used web-scraping and REST APIs to collect the data
- Perform data wrangling
  - We processed data with panadas and NumPy for cleaning and handling missing data
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - We build machine leaning models by using skit-learn for predictive analysis

Data Collection – SpaceX API

We used REST APIs and web-scraping for collecting the data from spaceX

https://github.com/Ammar-3A/IBM-Data-science-Capstone/blob/main/jupyter-labs-spacex-data-collection-api.ipynb

Request data from API

Convert result to Dataframe

Data cleaning and handle missing values

Data Collection - Scraping

- We use BeautifulSoup to convert HTML to pandas
- https://github.com/Ammar-3A/IBM-Data-science-Capstone/blob/main/jupyter-labs-webscraping.ipynb

Request URL and headers

Convert HTML to soup object

Going through rows to extract data

Data Wrangling

- We processed the data with pandas by seeing value types, counts, cleaning data and handling missing data.
- https://github.com/Ammar-3A/IBM-Data-science-Capstone/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb

Import data into a pandas dataframe

Process the data with pandas functions

The result will be a clean and filtered data

EDA with Data Visualization

- We used seaborn and matplotlib to visualize the data
- We use visualazations like scatter,bar,lineplot and others

- https://github.com/Ammar-3A/IBM-Data-science-Capstone/blob/main/jupyter-labs-eda-dataviz-v2.ipynb

EDA with SQL

- We used SQL query’s like SELECT, nested-query’s, functions like (MAX,MIN,COUNT)
- https://github.com/Ammar-3A/IBM-Data-science-Capstone/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

- We used markers and maps to show the locations of launch sites and highlight the near proximities around the site
- We added these markers to explain the sites in more details on the map

- https://github.com/Ammar-3A/IBM-Data-science-Capstone/blob/main/lab-jupyter-launch-site-location-v2.ipynb

Build a Dashboard with Plotly Dash

- We added a pie chart, scatter and range slider
- Pie chart will show the successes rate of all the sites and an individual one for each site
- Range slider to choose payload for the missile
- Scatter plot shows the success launch for each range of payload kg
- https://github.com/Ammar-3A/IBM-Data-science-Capstone/blob/main/Dash.txt

Predictive Analysis (Classification)

- We build ML models with skit-learn. First, we preprocessed the data, then we fitted a grid search for every model. Then we tweak the models with a validation set. Then we assess the models with a test set

Preprocessing data

Training models

Fine tuning models

Assessment of models

https://github.com/Ammar-3A/IBM-Data-science-Capstone/blob/main/SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb

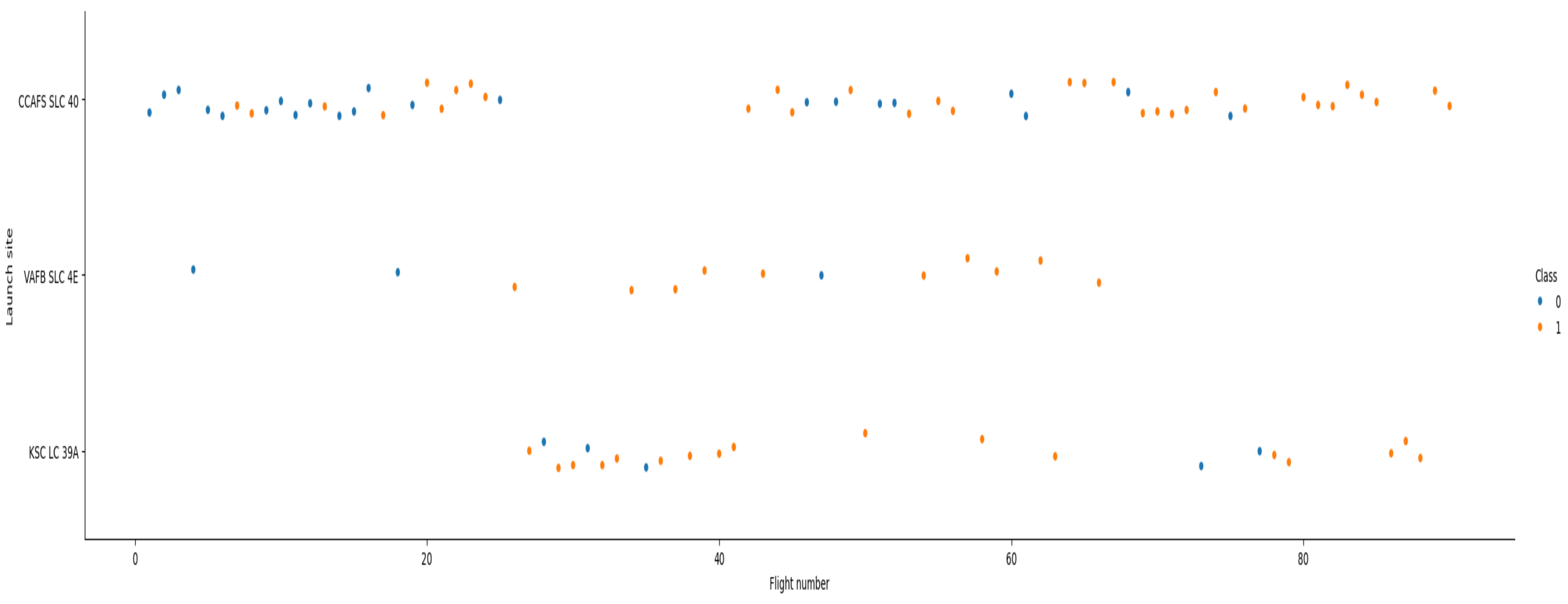
Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

Section 2

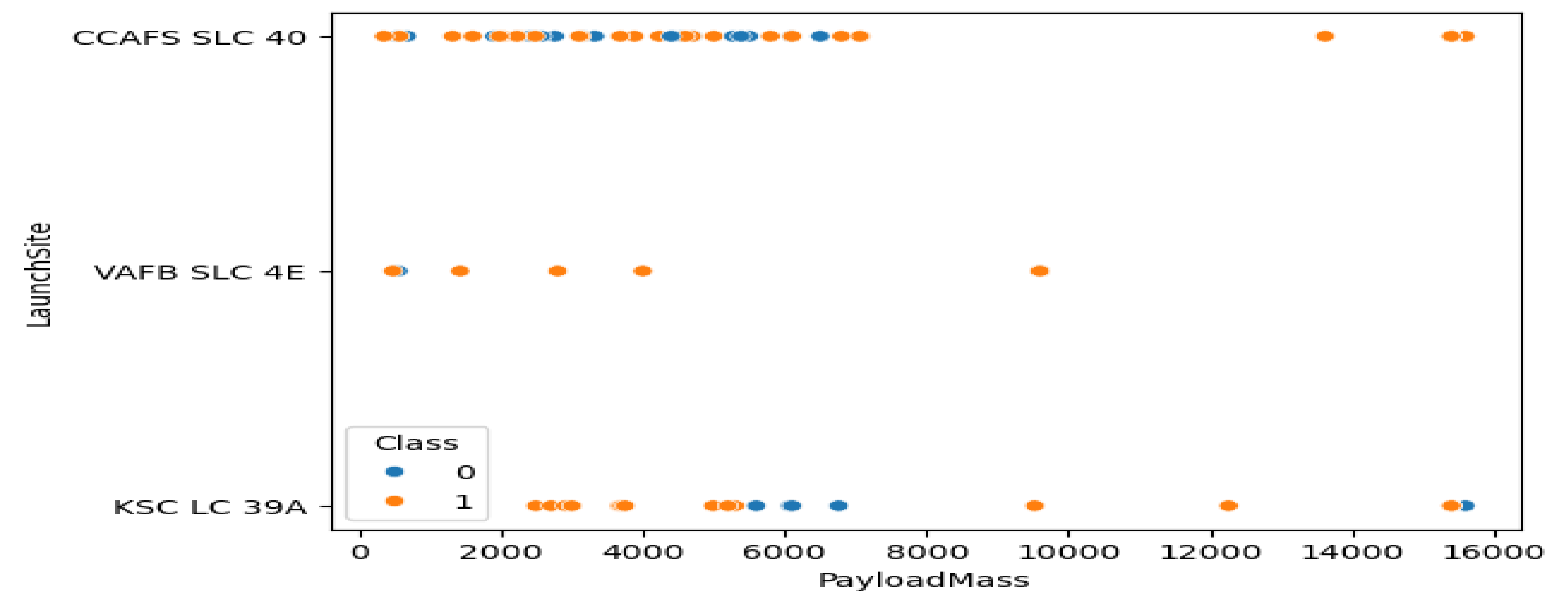
Insights drawn from EDA

Flight Number vs. Launch Site



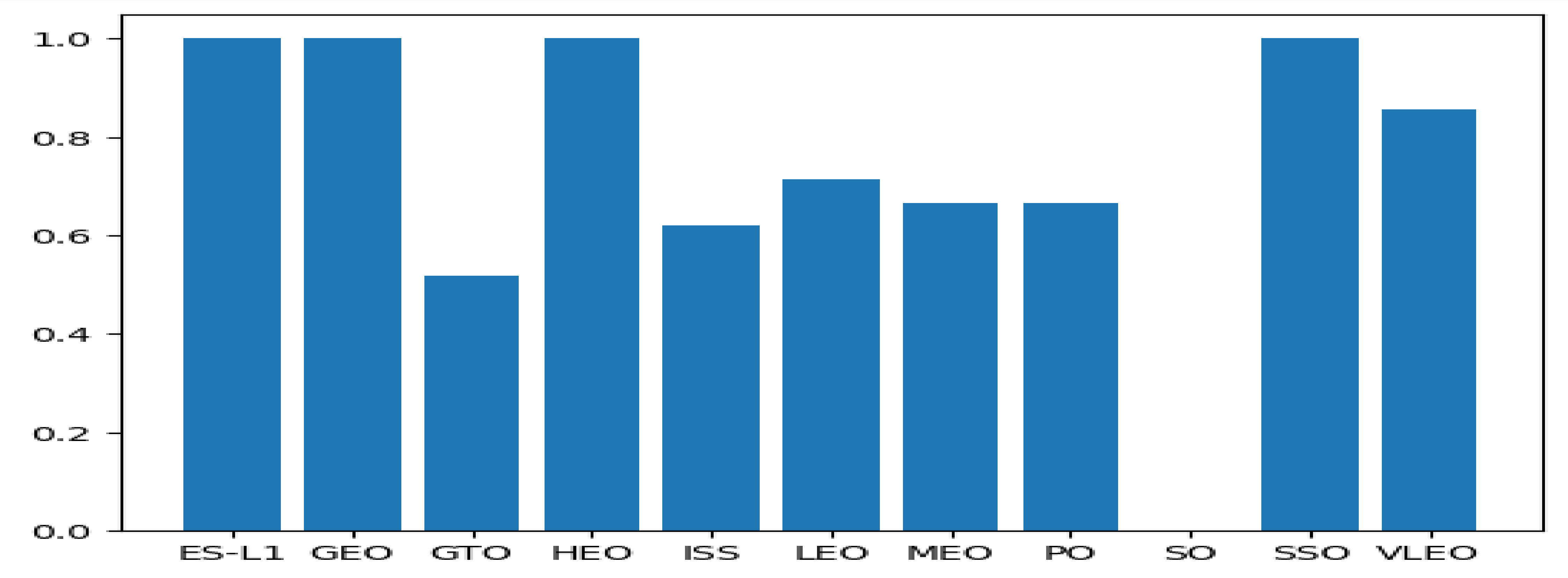
we notice with more flight numbers, the more likely the land will be a success

Payload vs. Launch Site



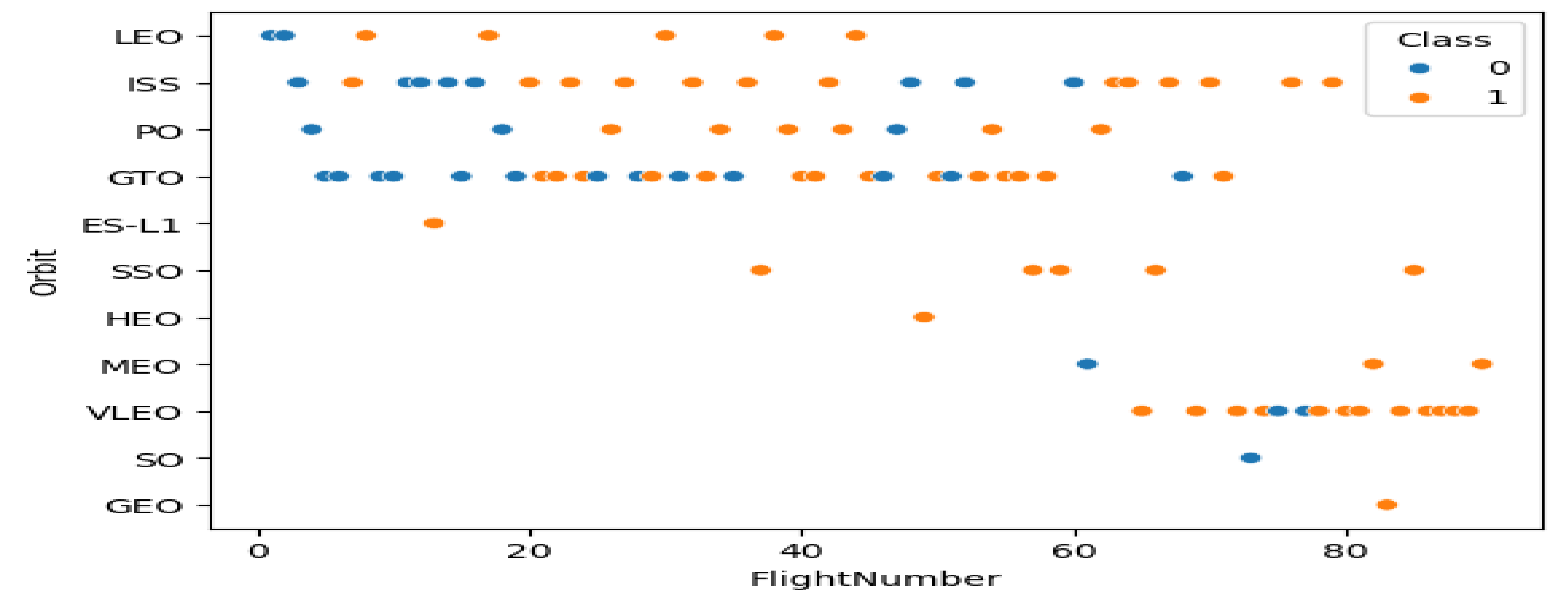
VAFB SLC 4E is not launching anything greater then 10000 kg
Abd CCAFS SLC 40 is more focused on the sub 8000 kg

Success Rate vs. Orbit Type



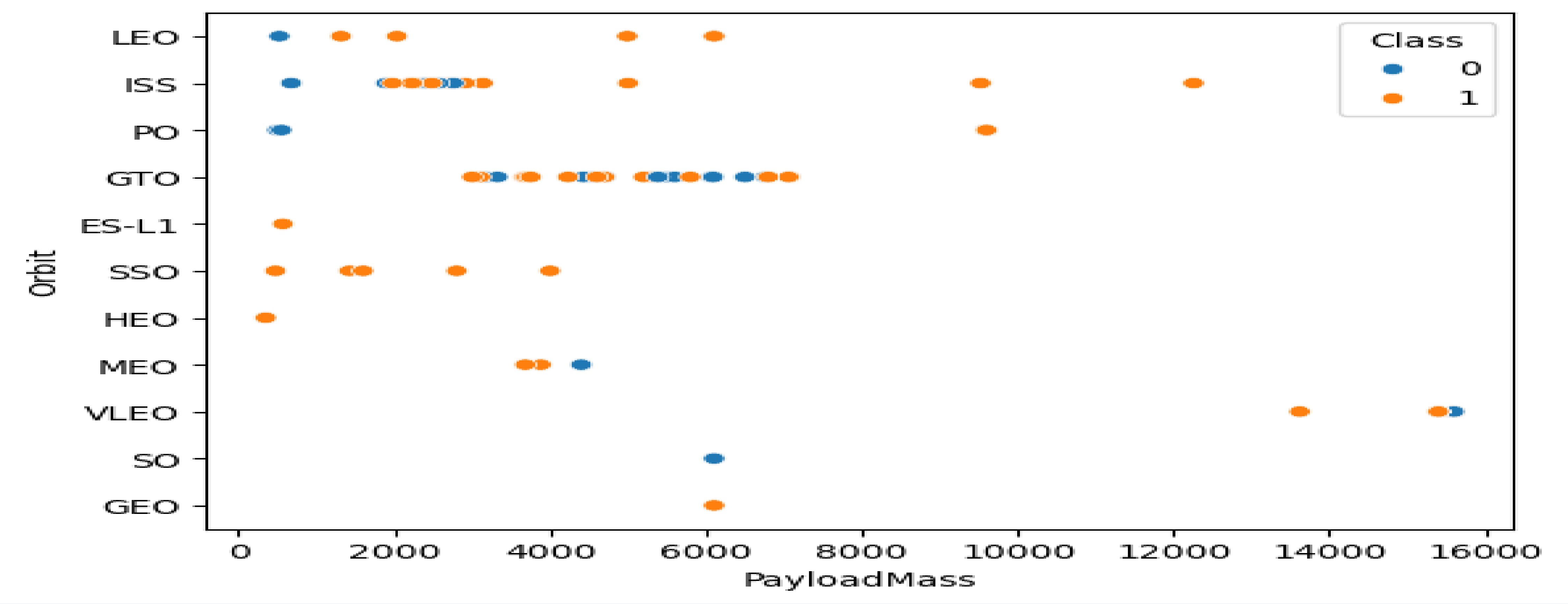
We see that ES-L1,GEO,HEO,SSO have the highest success rate of landing successfully

Flight Number vs. Orbit Type



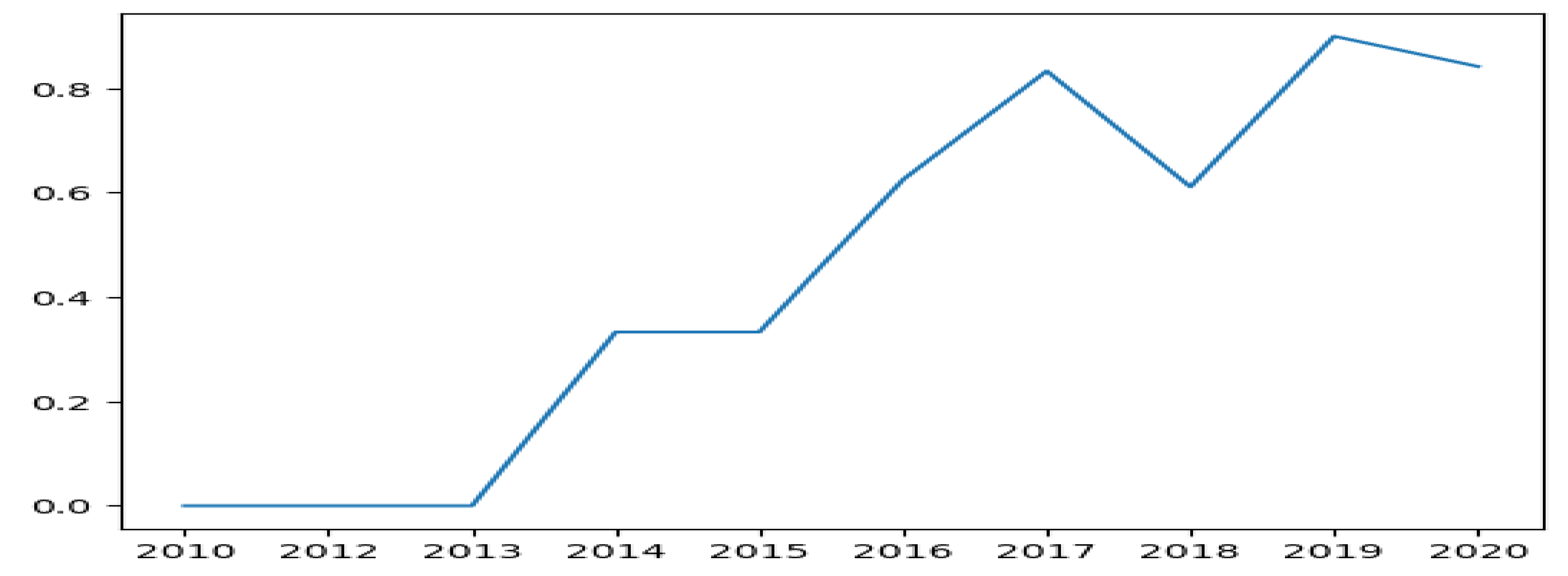
LEO has a relation with more Flight Numbers, the more highly the landing will be a success
While the others have no clear relation

Payload vs. Orbit Type



Here we see that for LEO,ISS,PO. The higher the payload mass, the more likely the landing will be a success
As for MEO, there's a limit in payload mass around 4000 and 15000 for VLEO
As for the others, there's no clear relation

Launch Success Yearly Trend



From 2013 to 2014, there's a noticeable increase. And there's a hold through 2014-2015. And after 2015, we see the trend continue with an increasing

All Launch Site Names

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Here we see the unique names of the launch sites

Launch Site Names Begin with 'CCA'

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-12-03	22:41:00	F9 v1.1	CCAFS LC-40	SES-8	3170	GTO	SES	Success	No attempt
2014-01-06	22:06:00	F9 v1.1	CCAFS LC-40	Thaicom 6	3325	GTO	Thaicom	Success	No attempt
2014-04-18	19:25:00	F9 v1.1	CCAFS LC-40	SpaceX CRS-3	2296	LEO (ISS)	NASA (CRS)	Success	Controlled (ocean)
2014-07-14	15:15:00	F9 v1.1	CCAFS LC-40	OG2 Mission 1 6 Orbcomm-OG2 satellites	1316	LEO	Orbcomm	Success	Controlled (ocean)
2014-08-05	8:00:00	F9 v1.1	CCAFS LC-40	AsiaSat 8	4535	GTO	AsiaSat	Success	No attempt
2014-	5:00:00	F9 v1.1 B1011	CCAFS LC-	AsiaSat 6	4428	GTO	AsiaSat	Success	No attempt

We see here all the records are shown here is launch_sites that start with 'CCA'

Total Payload Mass

Customer	total
NASA (CCD)	12055
NASA (CCDev)	12530
NASA (CCP)	12500
NASA (COTS)	525
NASA (COTS) NRO	0
NASA (CRS)	45596
NASA (CRS), Kacific 1	2617
NASA (CTS)	12050
NASA (LSP)	362
NASA (LSP) NOAA CNES	553
NASA / NOAA / ESA / EUMETSAT	1192

We see here there's a different sectors in NASA
These are the total for each sector

Average Payload Mass by F9 v1.1

Booster_Version	avg(PAYLOAD_MASS__KG_)
F9 v1.1	2928.4

The Average Payload of F9 v1.1 is around 2928kg

First Successful Ground Landing Date

Landing_Outcome	First landing
Success (ground pad)	2015-12-22

The first successful landing was in 2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

Booster_Version	PAYLOAD_MASS__KG_
F9 FT B1022	4696
F9 FT B1026	4600
F9 FT B1021.2	5300
F9 FT B1032.1	5300
F9 B4 B1040.1	4990
F9 FT B1031.2	5200
F9 B4 B1043.1	5000
F9 B5 B1046.2	5800
F9 B5 B1047.2	5300
F9 B5 B1048.3	4850
F9 B5 B1051.2	4200
F9 B5B1060.1	4311
F9 B5 B1058.2	5500
F9 B5B1062.1	4311

This is a sample of the boosters that landed successfully While having a payload in range 4000-6000

Total Number of Successful and Failure Mission Outcomes

Mission_Outcome	Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

We see here 98 successes
1 success with ground pad
1 success with unclear statue of payload
And 1 failure

Boosters Carried Maximum Payload

Booster_Version	PAYLOAD_MASS__KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

These boosters have the payload capped at the maximum limit Which is 15600

2015 Launch Records

month	Year	Landing_Outcome	Launch_Site	Booster_Version
01	2015	Failure (drone ship)	CCAFS LC-40	F9 v1.1 B1012
04	2015	Failure (drone ship)	CCAFS LC-40	F9 v1.1 B1015

We see here that there's two failed landings in 2015
One in January and one in April

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

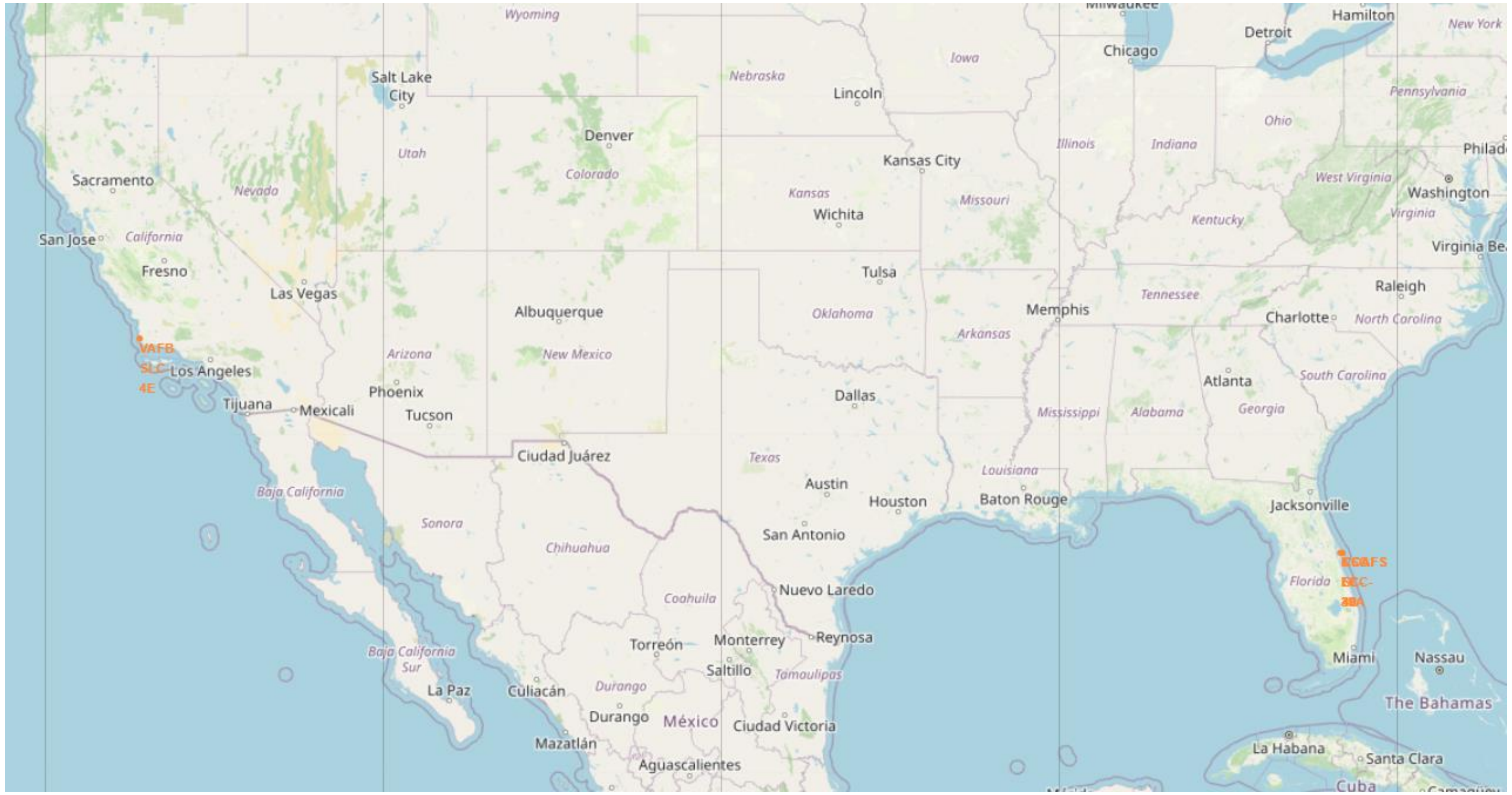
Landing_Outcome	outcomes	Date
No attempt	10	2012-05-22
Success (drone ship)	5	2016-04-08
Failure (drone ship)	5	2015-01-10
Success (ground pad)	3	2015-12-22
Controlled (ocean)	3	2014-04-18
Uncontrolled (ocean)	2	2013-09-29
Failure (parachute)	2	2010-06-04
Precluded (drone ship)	1	2015-06-28

We see that most successful attempts appear at the end of 2015 and after

Section 3

Launch Sites Proximities Analysis

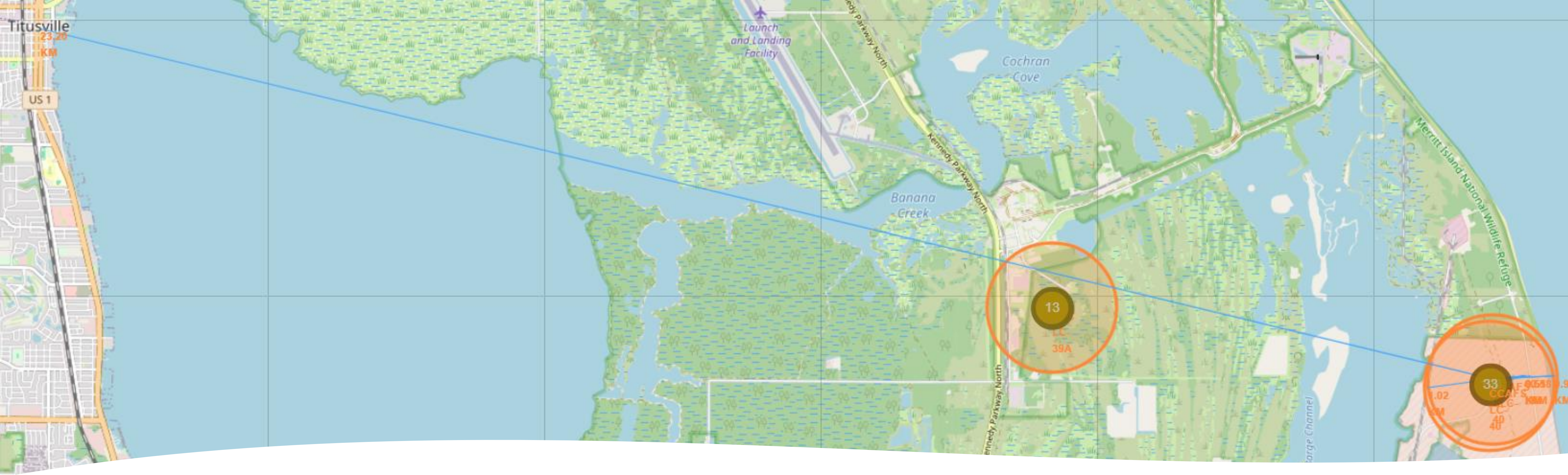
Launch site's locations



As we see here, there's four locations. 3 on the far east, and only one the far west

Example of labeled outcome of a launch site

- ***Here we take KSC LC-39A station as an example, we see that it had 10 successful landings, and 3 failures***

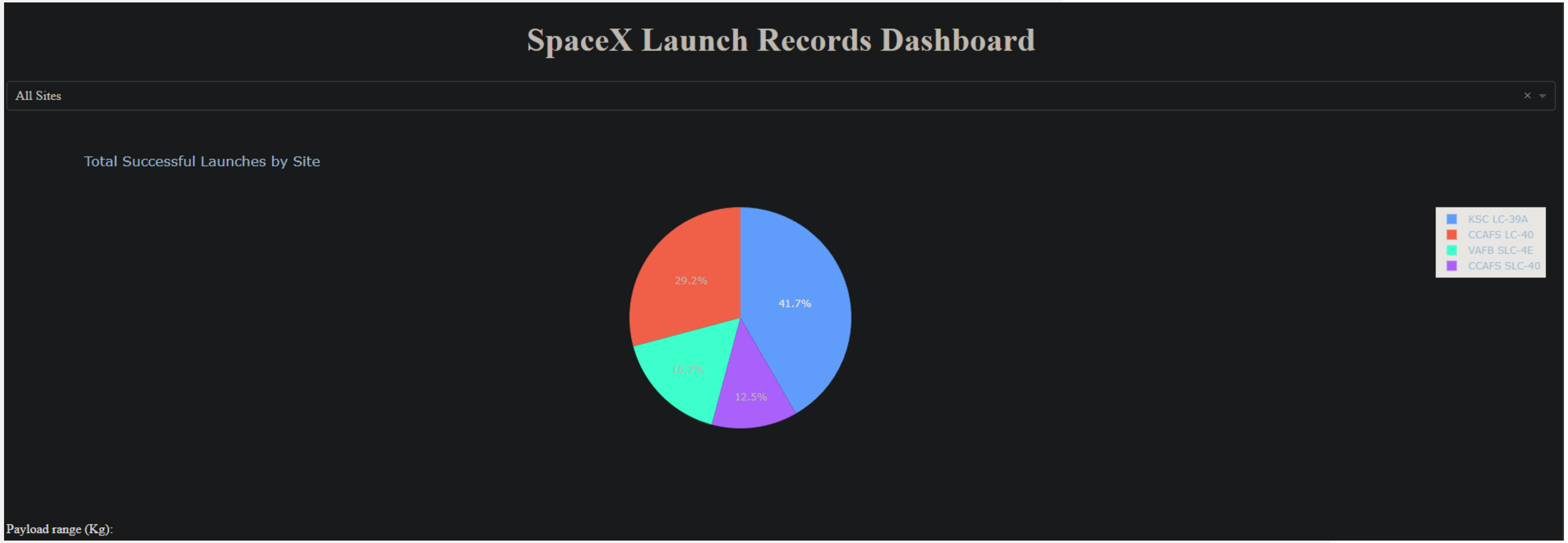


Distances between a launch site and its proximities

- ***Here we take the CCAFS SLC-40 station as an example, here we calculate the distance and draw a line for the nearest of the following:***
- ***A highway, a railroad , an ocean and a city***

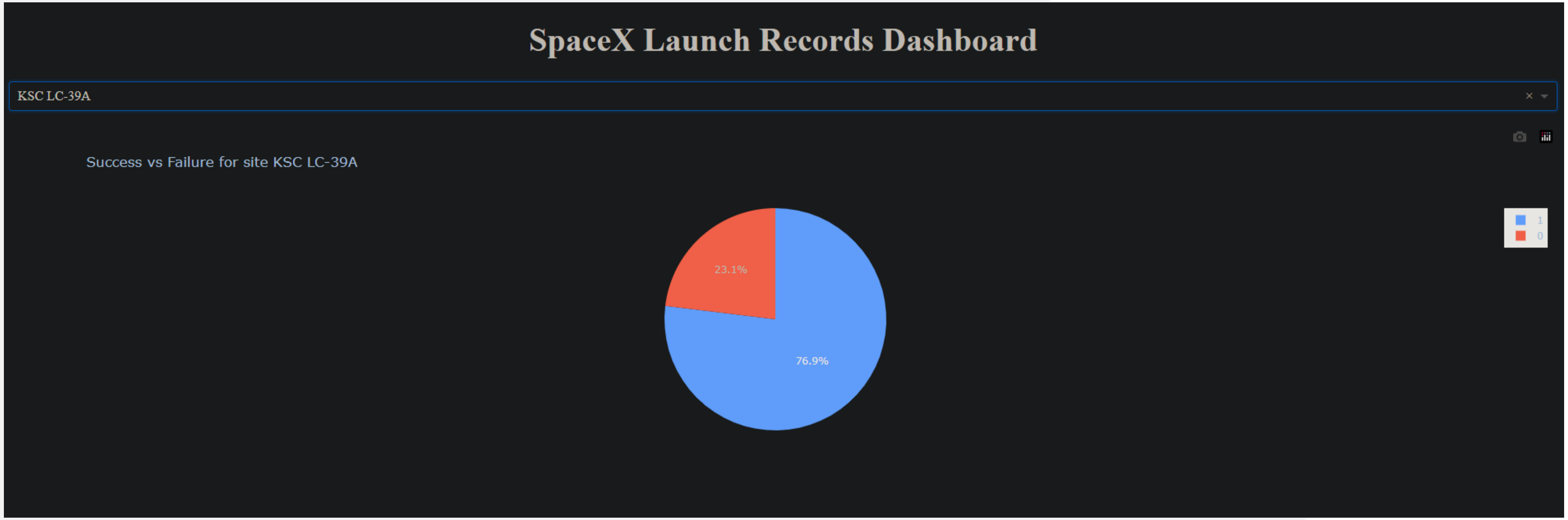
Section 4

Build a Dashboard with Plotly Dash



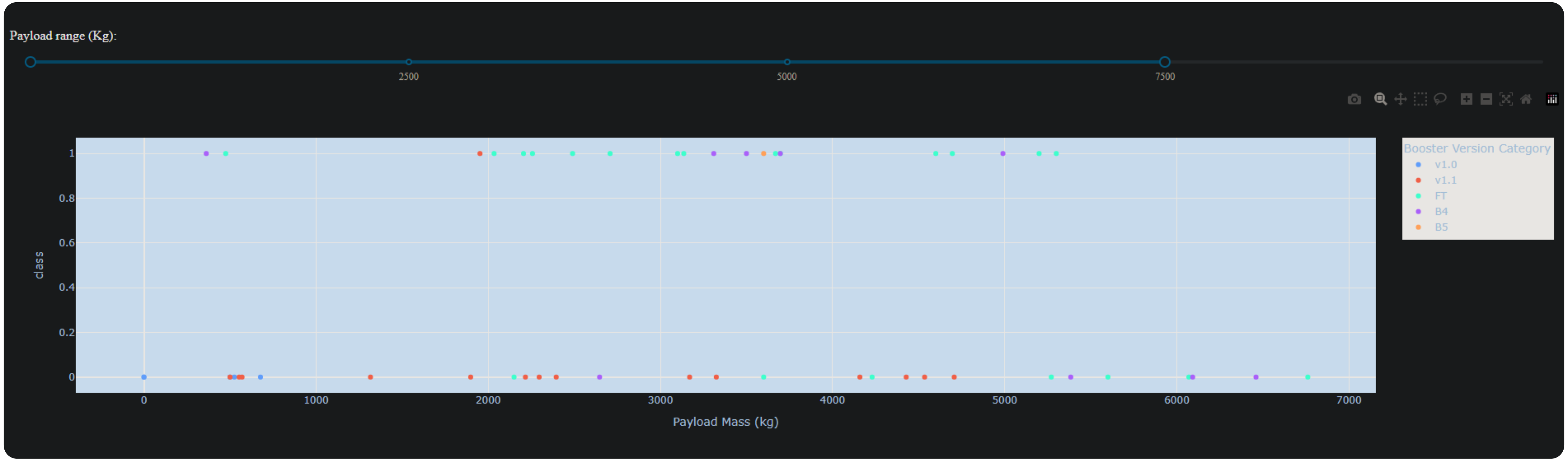
Total Successful Launches by Site

We see here that the KSC LC-39A has the most successful launches with a 41.7% over the other sites



Site with the highest Success ratio

The KSC LC-39A has a 76.9% success ratio, which is the highest ration between all sites

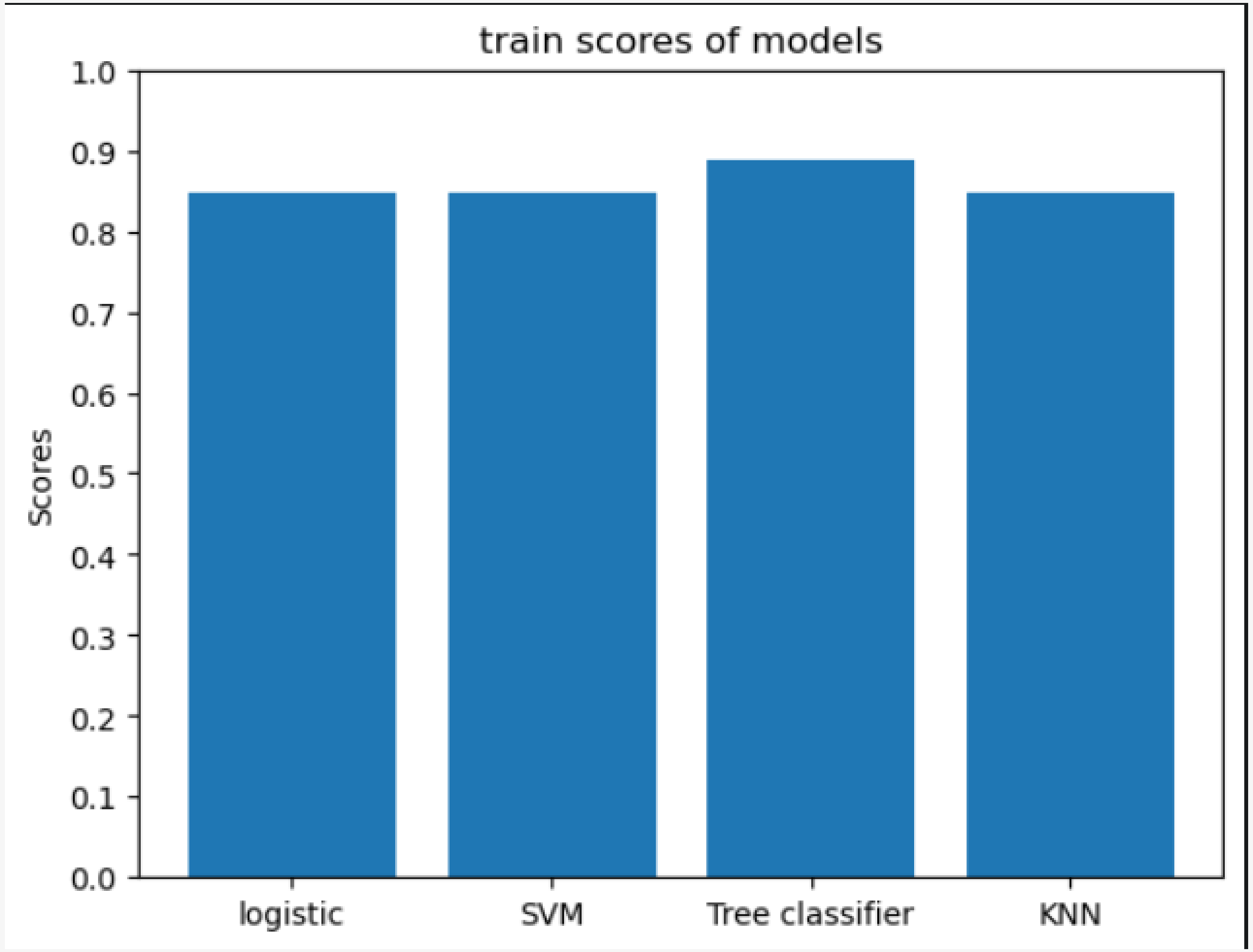


Successful launches over a range of payloads

- ***We selected here range from 0-7500 as an example, as we can see here that most successful launches are in the range***
- ***Of 2000-5000, and it's notable that the FT booster is the more frequent with success launches, especially in this range***

Section 5

Predictive Analysis (Classification)

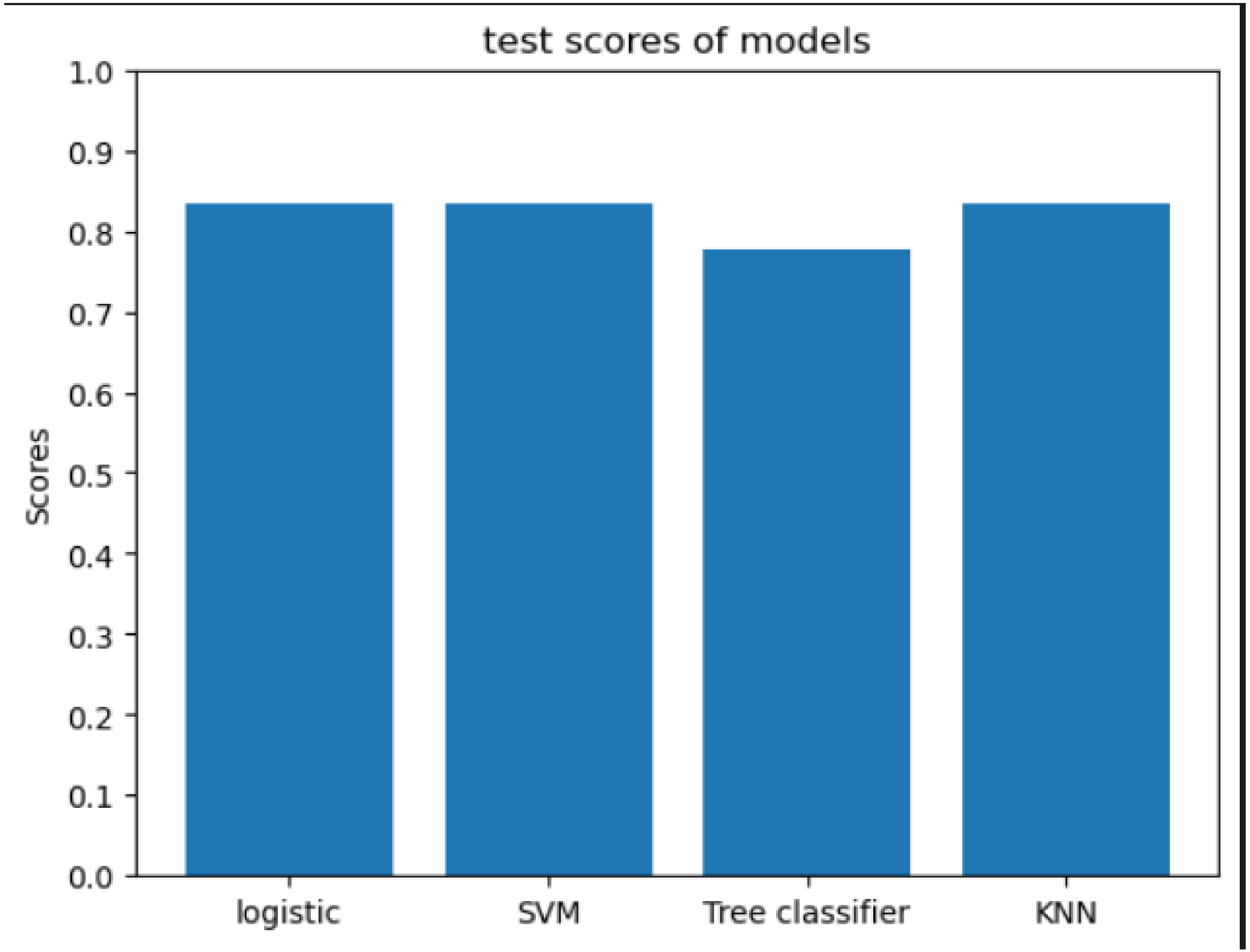


Classification Accuracy (for train)

- We notice that the Tree Classifier Has the highest score

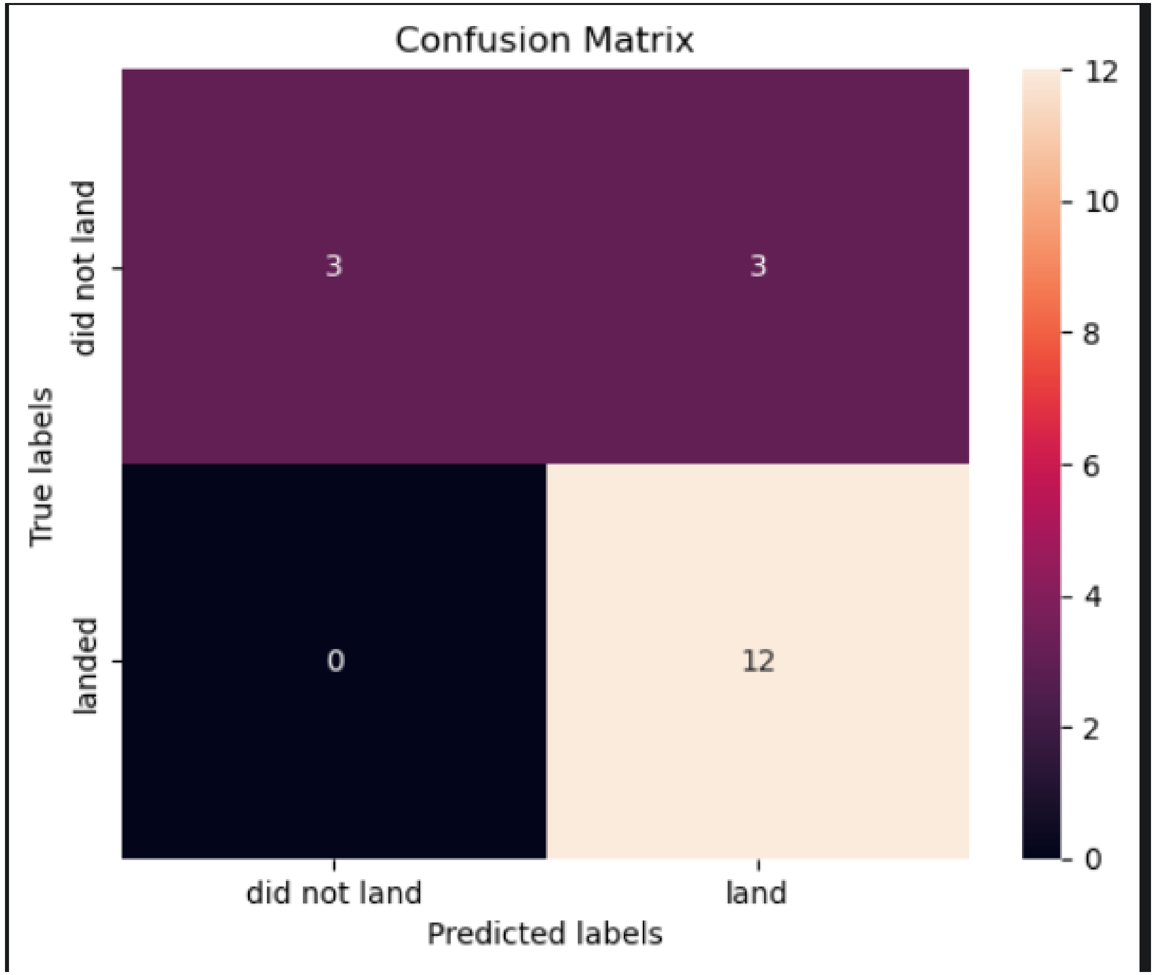
Classification Accuracy (for test)

- ***As we can see here, The KNN,SVM,logictic have the same score.***
- ***As for the Tree, it did poorly due to overfit***



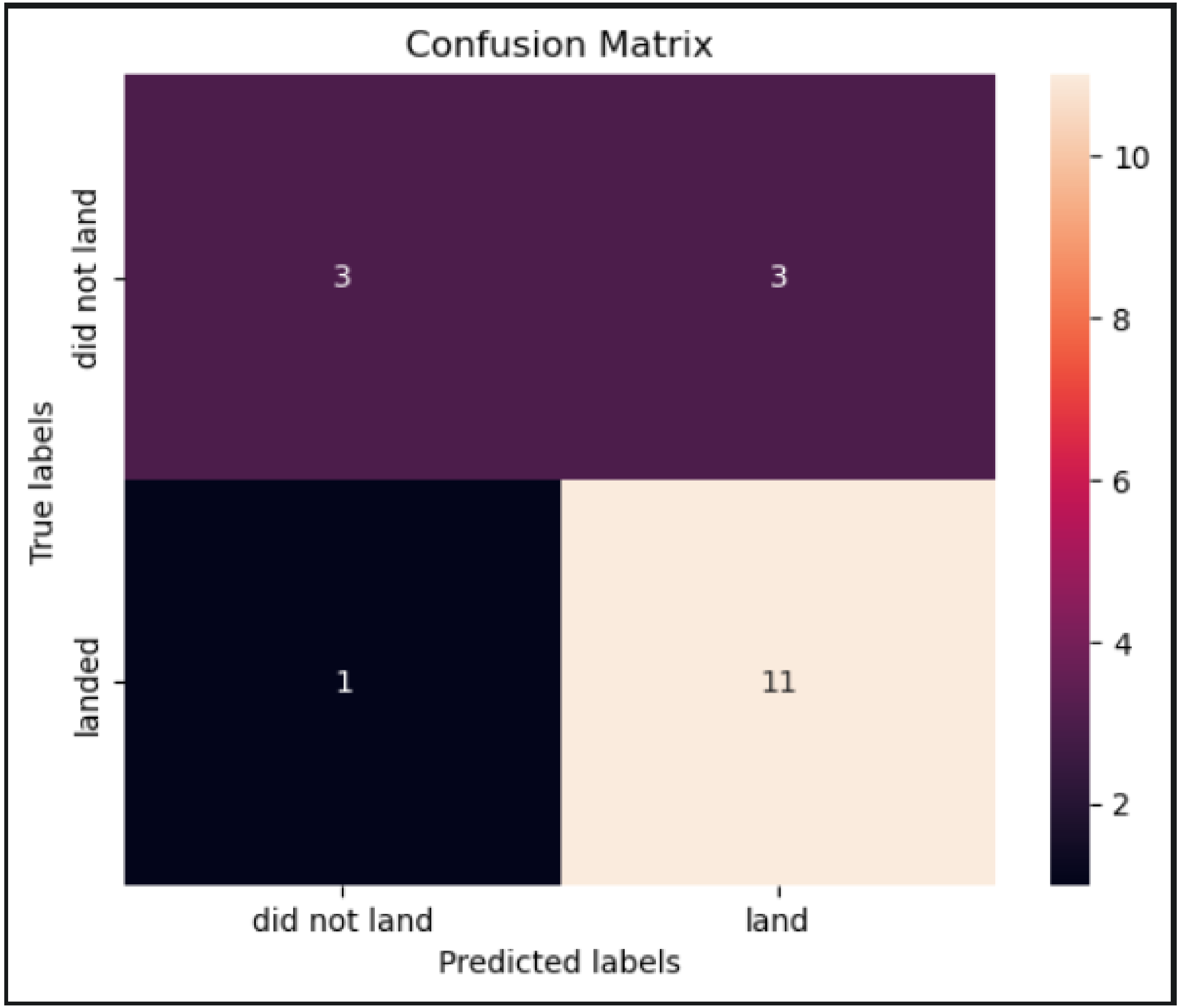
Confusion Matrix for best models

- ***This is the matrix for the KNN,logistic,SVM models.***
- ***Since they have the same Test scores. They all have the same matrix***
- ***It has 0 false negatives and 3 false positives***



Confusion Matrix for the worst model

- ***This is the matrix of the tree model.***
- ***As you can see it has 1 false negative and 3 false positive Which is worse than the other models***



Why the tree model did poorly ?

- The tree model is not consistent with results, since every time you train the model, the results vary due to different splits in the data.
- This problem is more impactful on smaller data sets

Conclusions

- we found that ES-L1,GEO,HEO,SSO Orbits are the most suitable for launching payloads
- We found that LEO,ISS,PO Orbit is suitable for high payload
- We found that KNN,SVM,Logistic models have similar performance, while Tree model have a vary one

Thank you!